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MANY OF THE GREATEST discoveries and most spectacular images of the universe have come from observatories in space. Above Earth's atmosphere, telescopes can see the sky far more clearly than those on the ground, and they can detect wavelengths that the atmosphere blocks.

VISIBLE AND ULTRAVIOLET LIGHT

Among the first successful space telescopes were those designed to detect ultraviolet light, notably NASA's Orbiting Astronomical Observatory series, launched between 1966 and 1972, and the International Ultraviolet Explorer, which was launched in 1978 and carried a 1.5 ft (0.45 m) telescope. Probably the most famous space telescope is the Hubble Space Telescope (HST), which was launched in 1990 and is still in operation. With a 7.9 ft (2.4 m) telescope designed primarily to detect visible and ultraviolet light, the HST has, among other successes, helped determine the age of the universe and produced evidence for the existence of dark energy. Space telescopes have also advanced more traditional realms of astronomy. For example, a European observatory called Gaia was launched in 2013 to measure the positions, motions, and distances of stars in the Milky Way, building up a three-dimensional picture of our galaxy.

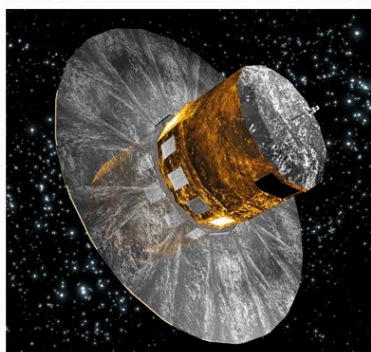
LAUNCH OF THE HUBBLE SPACE TELESCOPE

Launched in April 1990 from the Kennedy Space Center, Florida, on board the Space Shuttle Discovery, the Hubble Space Telescope orbits about 380 miles (600 km) above Earth. Initially intended to operate for 10 years, Hubble is still in operation thanks to five servicing missions by astronauts.



THE HUBBLE SPACE TELESCOPE

The entire Hubble craft is 43.5 ft (13.2 m) long and 14 ft (4.2 m) wide. The telescope is a reflector with a mirror 7.9 ft (2.4 m) in diameter. It operates primarily in visible light and ultraviolet, although its coverage also extends into the near-infrared.



GAIA ASTROMETRY SATELLITE

Launched at the end of 2013, the European Space Agency's Gaia spacecraft is mapping the precise positions and movements of over a billion stars throughout our galaxy in a mission scheduled to last until 2024.

HUBBLE DEEP-SKY VIEW

Thousands of galaxies extending into the remote depths of the universe are visible in this image from the Hubble Space Telescope. Called the Extreme Deep Field, it is the combined result of over 3 weeks of exposure time and covers an area less than a tenth of the width of the full Moon. The most distant galaxies are seen as they were over 13 billion years ago, when the universe was less than 5 percent of its present age. Many of them eventually developed into galaxies similar to our own Milky Way.

